Plan: speed, measurements, and next steps for stochastic SPH

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Abstract

The full Hamiltonian stochastic runs work (`fullruns.tex`). This note plans the next stage: where the time goes and how to cut it, running at NERSC, measuring correlators in thermalized ensembles, guarding against superluminal proposals, parallelizing the code, and the physics to do once all of this works.

I. WHERE THE TIME GOES

a. Profile. One time step of the coupled run (stochastic half step, leapfrog, stochastic half step), single thread, measured while other runs shared the machine, so absolute times are pessimistic:

	$h/a = 2$, $N = 672$	$h/a = 2.5$, $N = 1020$
one stochastic half step	20 ms	37 ms
rebuild stars (neighbours, comoving weights, 4×4 solve)	8	15
gradient correction (stars, fits, g_a twice)	3–15	5
stress sweep	3	6
residual sweep (single vote)	3	7
residual sweep (per-bond vote)	8	14
smoothing, thermodynamics, particle energies	4	5
leapfrog (ideal SPH)	5	12
one time step	$\simeq 45$ ms	$\simeq 85$ ms
per particle and step	$\simeq 65$ μs	$\simeq 85$ μs

About 90% of the time is in the stochastic step, and within it the Metropolis sweeps are only about a third. Most of the rest is bookkeeping: rebuilding the stars and the fits from scratch twice per step, with many small allocations. One equation-of-state inversion costs 0.8 μs.

b. Cost of a run. The cost is

$$\text{wall time} \simeq c \times N \times N_{\text{steps}}, \qquad c \simeq 65\ \mu\text{s (now, one thread)}. \tag{1}$$

The present test box ($N = 672$, $t = 100$, 11,500 steps) takes about 8 minutes per realization on one thread. A box of side $100h$ at $h/a = 2$ has $N \simeq 46{,}000$; to $t = 100$ it would take about 10 hours per realization on one thread now.

c. Options, cheapest first.

1. *Merge the two stochastic half steps.* The second half step of one time step and the first of the next act at the same positions. Doing one stochastic step of length Δt between leapfrogs (and half steps only at the ends of the run or at measurements) halves the stochastic cost. The proposals grow by $\sqrt{2}$, so the acceptance drops somewhat. Saving: about 45%.

2. *Build the stars once per step and reuse the neighbour lists.* Keep a neighbour list with a skin and rebuild it only when a particle has moved more than half the skin; recompute only the weights. Share the stars between the bond weights and the gradient fit. Saving: most of the star cost.

3. *Remove allocations in the hot loops:* preallocated buffers, static 2×2 fits in closed form instead of `pinv`, arrays of structures replaced by flat arrays. Typical gain in Julia: a factor 2–5.

4. *Single vote* instead of the per-bond vote: half the residual cost and higher acceptance.

5. *Threads* (Sect. V) and *ensembles* (Sect. II).

Items 1–4 together should bring c to roughly 10 μs per particle and step on one thread, a factor 5–7.

II. RUNNING AT NERSC

The CPU nodes of Perlmutter have 128 cores (two 64-core AMD EPYC sockets). Julia is available as a module, and the package environment of the repository (`Project.toml`, `Manifest.toml`) can be instantiated in the user area; PyPlot is not needed on the compute nodes (plots are made afterwards).

- *Ensembles first.* Correlators need many independent realizations. These are embarrassingly parallel: one realization per group of threads, many groups per node, many nodes through a job array. This needs no change to the code beyond writing each realization's output to its own file.

- *Threads within a realization* for large boxes (Sect. V).

- *Domain decomposition* across nodes only if single realizations no longer fit on one node; not needed soon.

Practical steps: install and precompile the environment, a small driver script that reads a parameter file and a seed, writes HDF5 (or JLD2) output, and can restart from a checkpoint; a Slurm job-array template.

III. MEASUREMENTS IN THERMALIZED ENSEMBLES

a. Start in equilibrium. The slowest modes relax on the diffusion time of the box, $1/(\nu k_{\min}^2)$ with $\nu = \eta/w$ and $k_{\min} = 2\pi/L$. For the present parameters $\nu = 2.2 \times 10^{-3}$; this

is about 4 time units for $L = 0.6$ but about 300 for $L = 5$. Waiting for thermalization from rest is therefore not an option for large boxes. Instead, start from an equilibrium draw: momenta drawn from the single-particle covariance (`stochrep.tex`, the moving fluid), with the total momentum fixed, and a short burn-in for the particle positions and the lattice modes.

b. What to record. For a fixed set of wave vectors (all $\boldsymbol{k}$ with $|\boldsymbol{k}|$ below a cut, say $kh < 1.5$), record at regular intervals the particle sums

$$\boldsymbol{M}_{\boldsymbol{k}}(t) = \sum_a \boldsymbol{m}_a(t)\, e^{-i\boldsymbol{k}\cdot\boldsymbol{x}_a(t)}, \qquad S_{\boldsymbol{k}}(t) = \sum_a s_a(t)\, e^{-i\boldsymbol{k}\cdot\boldsymbol{x}_a(t)}, \qquad E_{\boldsymbol{k}}(t) = \sum_a e_a(t)\, e^{-i\boldsymbol{k}\cdot\boldsymbol{x}_a(t)}, \tag{2}$$

the Fourier transforms of the momentum, entropy and energy densities. For $kh \ll 1$ they coincide with the transforms of the smoothed fields. Split the momentum into its longitudinal and transverse parts,

$$M^L_{\boldsymbol{k}} = \hat{\boldsymbol{k}} \cdot \boldsymbol{M}_{\boldsymbol{k}}, \qquad M^T_{\boldsymbol{k}} = \hat{\boldsymbol{k}} \times \boldsymbol{M}_{\boldsymbol{k}}. \tag{3}$$

This is a cost of order $N \times N_k$ per record, small if recorded every few steps. Store the time series per realization (HDF5), with the energies, the acceptance and the parameters.

c. The transverse correlator. For example, with $\boldsymbol{k}$ along x,

$$C_T(k,\tau) = \frac{1}{V}\,\langle M_y(t+\tau,\boldsymbol{k})\, M_y^*(t,\boldsymbol{k})\rangle \tag{4}$$

(the asked-for $\langle m_y(t,k)\, m_y(0,-k)\rangle$, since $M_{-\boldsymbol{k}} = M^*_{\boldsymbol{k}}$). Estimate it by averaging over the start time t in the stationary state (by FFT of the time series, or directly over lags), over all $\boldsymbol{k}$ of the same length (with M_y replaced by M^T), and over realizations. Errors from the spread over realizations, or by jackknife over time blocks longer than the correlation time. Prediction from linearized hydrodynamics:

$$C_T(k,\tau) = \chi_T\, e^{-\nu k^2|\tau|}, \qquad \chi_T = wT, \tag{5}$$

so the equal-time value checks the statics and the decay rate measures the shear viscosity at that k, including its renormalization by the fluctuations.

d. The longitudinal correlator. $C_L(k,\tau)$ from M^L: two damped sound modes,

$$C_L(k,\tau) \simeq \chi_L\, e^{-\frac{1}{2}\Gamma_s k^2|\tau|}\cos(c_s k\tau), \qquad \Gamma_s = \frac{\frac{4}{3}\eta + \zeta}{w} \quad \text{(3D fluid, planar flow)}, \tag{6}$$

plus a small non-oscillating correction. Together with $\langle E_{\boldsymbol{k}} E^*_{\boldsymbol{k}}\rangle$, $\langle S_{\boldsymbol{k}} S^*_{\boldsymbol{k}}\rangle$ and their cross correlators this gives the full set of static susceptibilities and the sound speed and attenuation.

e. Errors. Two levels. Realizations are independent, so the mean and the standard error over realizations are the most robust estimate. Within a realization the time series is correlated over the decay time $1/(\nu k^2)$, which is long at small k; binning the series into blocks longer than this (and measuring the integrated autocorrelation time) gives honest errors from few long runs. Derived quantities (a fitted $\nu(k)$, ratios of correlators) get errors by jackknife over realizations or time bins. Julia packages: `Statistics` (standard library), `BinningAnalysis.jl` (binning, jackknife, autocorrelation times for Monte Carlo time series), `MCMCDiagnosticTools.jl` (effective sample size, agreement between independent chains), `LsqFit.jl` (fits with parameter covariances), `FFTW.jl` (correlators by FFT), `Bootstrap.jl`.

f. Tools to build.

- An observer hook in the time loop: a list of measurements, each called every n steps, writing to the output file.
- The equilibrium initial state.
- A post-processing module: read the files, correlators by FFT, averages over $|\boldsymbol{k}|$ shells and realizations, errors, fits of the decay rates, and comparison with the predictions above.
- Spectra $\langle |M_{\boldsymbol{k}}^{L,T}|^2 \rangle$ versus k as the first check of every run.

IV. GUARDING AGAINST SUPERLUMINAL PROPOSALS

A proposal can push a particle's momentum beyond what its energy allows. The inversion of the equation of state then fails (or gives $|v| \geq 1$). The fix is simple and exact: such a state has zero weight under $e^{\mathbb{S}}$, so the move is rejected. In practice

- in the vote, a failed or superluminal inversion for any member of the star counts as $\Delta\mathbb{S} = -\infty$, and the whole move is rejected;
- in the deterministic residual ($\lambda = \infty$) and in the back map, a superluminal result is clipped or the step is subdivided, and counted;
- counters for these events are written to the output.

Detailed balance is unaffected, since the reverse of a rejected move is never proposed from an allowed state. This is a few lines in `ask_star!` and should be done before the long production runs.

V. PARALLELIZATION

a. Do not start from scratch. The algorithm is now validated, and the profile shows that the cost is dominated by avoidable bookkeeping, not by the structure of the method. A rewrite in another language would cost months and the validation would have to be repeated. The plan is to refactor the hot paths in Julia, keeping the present code as the reference that every refactored routine must reproduce (same random streams, same results).

b. Steps.

1. A benchmark script with the per-part timing above, run before and after every change; Julia's profiler and allocation tracker on the hot loops.
2. The single-thread items of Sect. I.
3. Threads: the smoothing, thermodynamics, velocities and leapfrog loops are per particle and thread trivially. The stochastic sweeps already run the cells of one colour in parallel; with many cells (large boxes) this scales, with few cells (the present test box has 30) it does not, and ensembles are the better use of the cores.

4. Later, if needed: domain decomposition with a halo of width $2h$, which the checkerboard of cells already respects.

The random streams are per particle, so the results do not depend on the number of threads; this is the check that each step preserves.

VI. NEXT STEPS, ONCE THIS WORKS

1. *Equilibrium validation.* Static susceptibilities, C_T and C_L versus k and τ, in boxes large enough to resolve a range of $kh \ll 1$. The decay rates give η and Γ_s as functions of k: their approach to the input values as $k \to 0$, and the fluctuation correction.

2. *Renormalization of the viscosity.* In two dimensions the fluctuation correction to η grows logarithmically at small k (long-time tails). Measure $\eta(k)$ over a decade in k and compare with the one-loop prediction, including the cutoff dependence on h.

3. *Dependence on the discretization:* h/a, Δt, λ, and the single versus per-bond vote, for the measured transport coefficients.

4. *Energy drift.* Decide between a global thermostat and a fix of the Itô term of the change of representation (`fullruns.tex`).

5. *Moving and inhomogeneous backgrounds* with the Hamiltonian step on: a uniformly moving fluid (Lorentz covariance of the correlators), then shear flow and an expanding (Bjorken-like) background.

6. *Physics targets:* long-time tails and the renormalization of the transport coefficients in relativistic fluids, fluctuations near a critical point, and the density-frame formulation that motivated the scheme.

VII. STATUS

On the branch `nextstage`:

- *Superluminal guard*: done (`test/teststochrep_guard.jl`). A failed or superluminal inversion rejects the move; an unphysical intermediate state of the bond split falls back to the even split; the gradient back map falls back to the plain map; the deterministic residual subdivides its step. Counters are reported.

- *Merged stochastic half steps*: done in the new driver (`examples/stochrep_equilibrium.jl`); about half the cost per step. The residual acceptance drops from 0.88 to 0.77 (single vote).

- *Equilibrium initial state, recorder, correlators*: first version (`src/measure.jl`, `examples/stochrep_c`

- *Static solves* in the comoving weights and the gradient fit: done, no measurable speed-up; a singular moment matrix (very fast flow, $\gamma \gtrsim 2$, when too few bonds remain inside the support along the flow) now falls back instead of failing. The comoving stencil at large γ needs a closer look.

- *Next*: a proper profile (sampling profiler and allocation tracking) of `build_stars` and the gradient correction.

VIII. ORDER OF WORK

1. The superluminal guard (Sect. IV). Small; do first.
2. Merge the stochastic half steps; build the stars once per step. Check that the results are unchanged in distribution.
3. Equilibrium initial state, observer hook, HDF5 output, correlator post-processing; first measurement of C_T in the present box.
4. Allocation and fit clean-up; threads; benchmark.
5. NERSC: environment, driver, job-array template; first ensemble in a larger box.
6. The physics of Sect. VI.